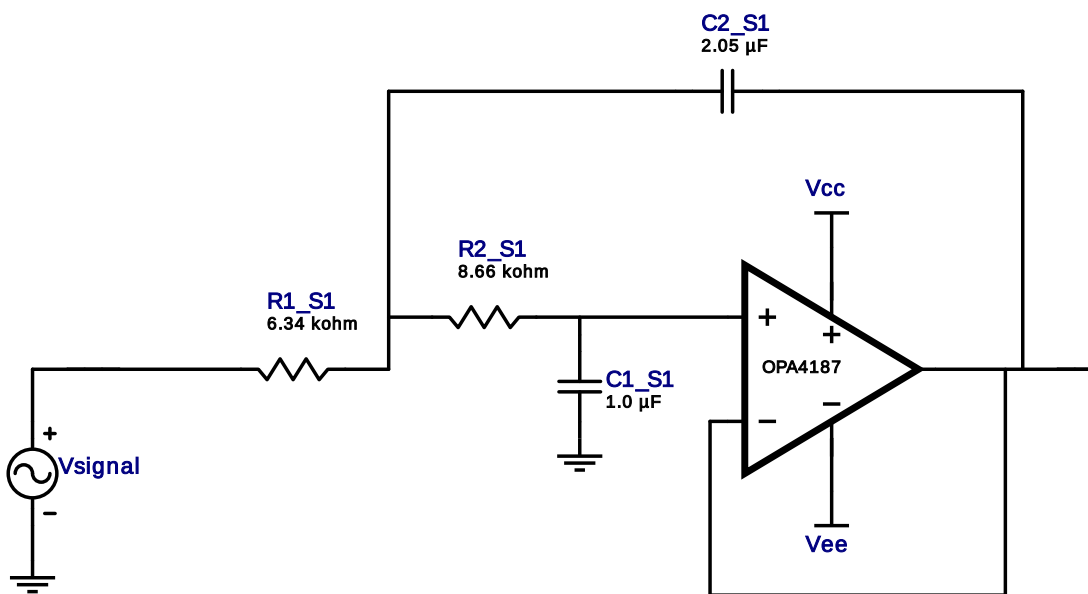


Filter Design Report

Design : Lowpass Filter - 2nd order Butterworth
Design ID: 8

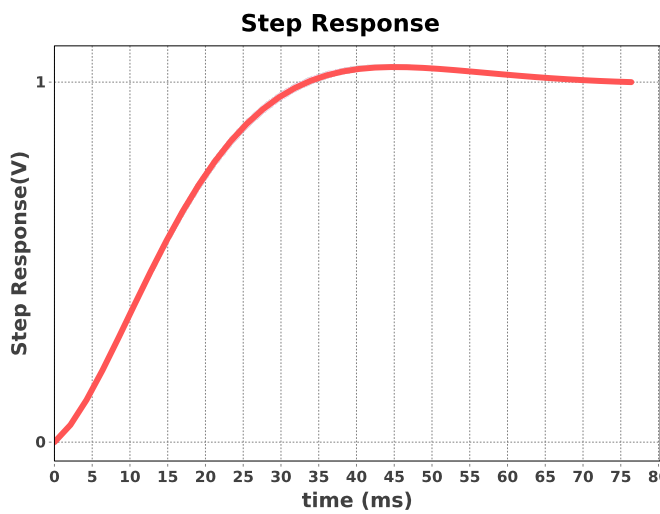
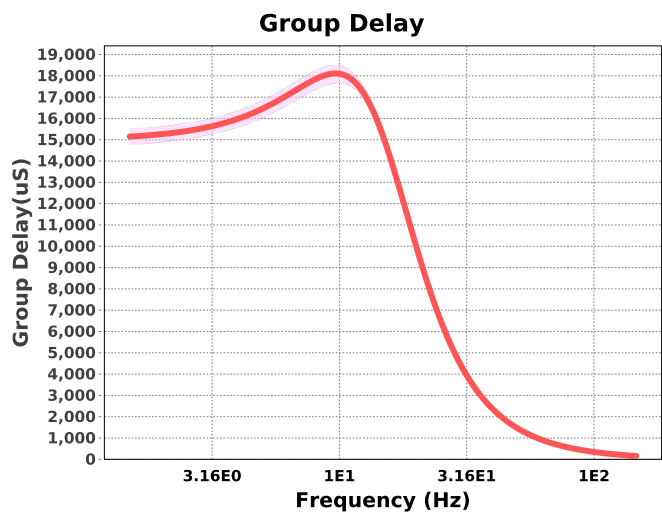
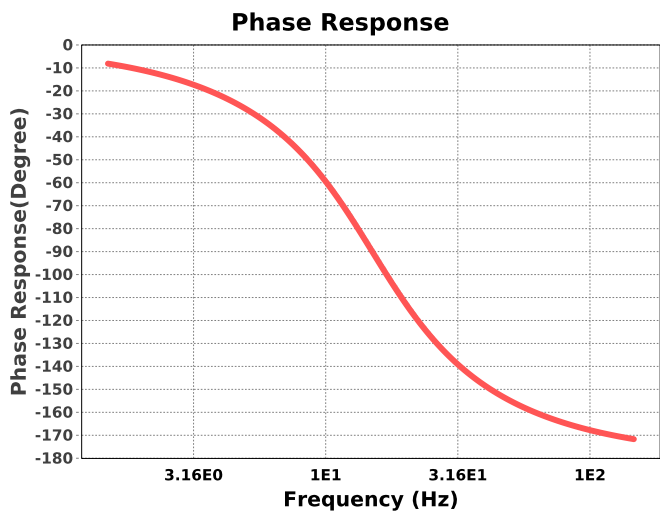
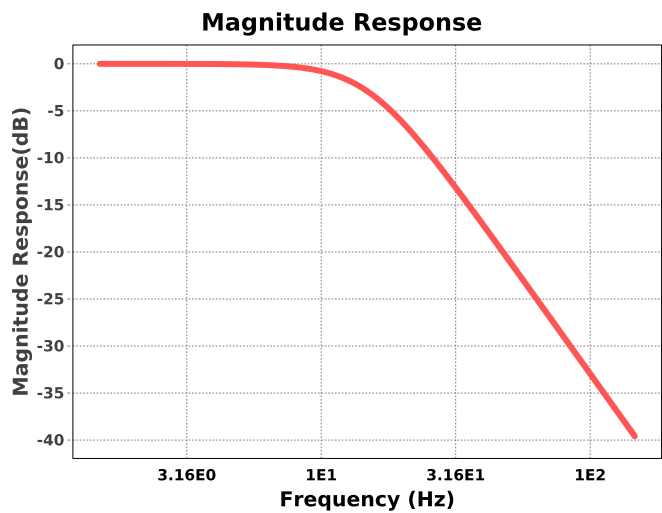


Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA4187	GbwTyp= 0.55MHz VccMax= 36V VccMin= 4.5V	1
2.	C1_S1	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 2.05 uF Tolerance= 2.0 %	1
4.	R1_S1	Generic	Ideal	Res= 6340.0ohm Tolerance= 1%	1
5.	R2_S1	Generic	Ideal	Res= 8660.0ohm Tolerance= 1%	1

Sensitivity Analysis

#	Name	Series	Tolerance
1.	Cap	E48	2%
2.	Res	E96	1%



Design Inputs

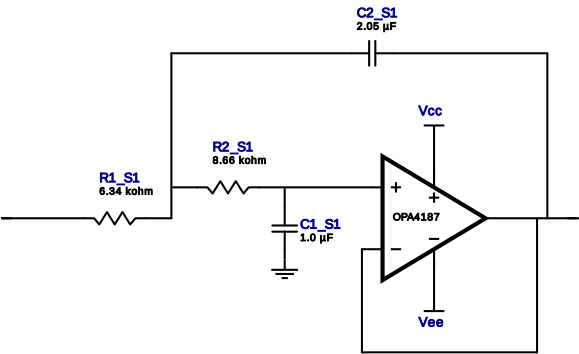
#	Name	Value	Description
1.	FilterType	lowpass	
2.	FilterResponse	Butterworth	
3.	FilterOrder	2.0	
4.	FilterTopology	Sallen-Key	
5.	NumberOfStages	1.0	
6.	PassbandFrequency	15.0	
7.	StopbandAttenuation	-20.95	
8.	StopbandFrequency	50.0	
9.	Gain	1.0	
10.	DualSupply	+/-5.00 V	Power supply(s) to active chips
11.	ResistorTolerance	E96	Resistor series - 1% Passive resistor tolerance
12.	CapacitorTolerance	E48	Capacitor series - 2% Passive capacitor tolerance

Design Assistance

1. **OPA4187** Product Folder : <http://www.ti.com/product/OPA4187> : contains the data sheet and other resources.

Filter Stage :1

Cutoff Frequency 15.002 Hz
Min GBW Reqd 1.061 kHz
Stage Gain 1.0 V/V
Stage Q 707.277 m
Stage Topology Sallen-Key



Electrical BOM

#	Name	Manufacturer	Part Number	Properties	Qty
1.	A1_S1	Texas Instruments Inc.	OPA4187	GbwTyp= 0.55MHz VccMax= 36V VccMin= 4.5V	1
2.	C1_S1	Generic	Ideal	Cap= 1.0 uF Tolerance= 2.0 %	1
3.	C2_S1	Generic	Ideal	Cap= 2.05 uF Tolerance= 2.0 %	1
4.	R1_S1	Generic	Ideal	Res= 6340.0ohm Tolerance= 1%	1
5.	R2_S1	Generic	Ideal	Res= 8660.0ohm Tolerance= 1%	1

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